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PI Name:	Barcellos-Hoff, Mary Helen Ph.D.		
Project Title:	NSCOR: Mechanisms of HZE Damage and Repair in Human Epithelial Cells		
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Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:	Dr. Andrew Wyrobek has joined the team to conduct microarray analysis. Dr. Aloke Chatterjee has retired, and is now acting in a advisory role. Dr. Sylvain Costes has assumed responsibility for modeling radiation effects.		
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Task Description:	NASA's Biomedical Critical Path Roadmap defines the carcinogenic risks of radiation exposure as one of only four Type I risks identified. Estimates of cancer risk from human travel in space should be based on a mechanistic understanding of complex effects that different types of radiation exposure elicit. Because most adult human solid tumors are epithelial in origin and because radiation effects are dictated by cellular genotype and phenotype, studies of HZE particle radiation biology using a physiologically relevant model will be the most informative regarding risk. We have developed methods for culturing human mammary epithelial cells (HMEC) derived from normal tissue as both monolayer and multicellular aggregates, which enable study of phenotypes and pathways that may be uniquely human, epithelial or multicellular. We propose to study how controlled cellular and microenvironment conditions affect early responses to HZE particle radiation-induced DNA damage in terms of repair, foci formation and gene expression, and persistent radiation-induced phenotypes and pathways that precede neoplasia. The potential for inducing genomic instability, which may contribute to its carcinogenic effect, will be measured in both finite lifespan HMEC and immortal HMEC. Together, these surrogate endpoints will enable functional evaluation of neoplastic potential as a function of radiation quality, energy and dose. These data will be integrated at two levels: by theoretical modeling of the physical events leading to DNA damage and by systems biology modeling of critical pathways. The intent of this 5-year project is to provide a comprehensive examination of HZE effects from the initial damage, to early cellular HMEC responses, to persistent functional precursors of carcinogenesis. We will address the mechanisms, nature, and frequency of DNA damaging events; mechanisms of DNA repair and misrepair; early signal transduction mechanisms; immediate and long-term, reversible and irreversible gene expression changes; cellular remodeling and reorganization; potential mechanisms of tissue "repair" and matrix effects; and the cellular and molecular mechanisms of HZE/proton-induced progression to a neoplastic phenotype. To provide a comprehensive picture of HZE effects from the initial damage, to early cellular HMEC responses, to persistent functional precursors of carcinogenesis. Several NASA areas of interest will be addressed: mechanisms, nature, and frequency of DNA damaging events; mechanisms of DNA repair and misrepair; early signal transduction mechanisms; immediate and long-term, and reversible and irreversible gene expression changes; cellular remodeling and reorganization; potential mechanisms of tissue "repair" and matrix effects; induction and regulation of genomic instability; and the cellular and molecular mechanisms of charged particle-induced progression to a neoplastic phenotype. The data obtained in these studies will be integrated at two levels: by theoretical modeling of the physical events leading to DNA damage and by systems biology modeling of critical pathways. Ultimately, this work will help NASA develop a basic understanding of the mechanisms of HZE radiation damage and repair and their contributions to the neoplastic process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Current modeling of risk relies on comparing the relatively biological effectivenesslow LET radiation to high LET radiation effects, thus providing an earth benefit in the description and quantitation of low dose reference radiation effects. In the context of the NSCOR, we have developed image analysis programs to quantify DNA damage response (e.g. radiation-induced foci) and specific organelles (i.e. centrosomes) using high-throughput imaging bioinformatics.
	LBNL NSCOR NASA TASK BOOK 2006 Mechanisms of HZE Damage and Repair in Human Epithelial Cells PI: Barcellos-Hoff Co-PI: Chatterjee, Costes, Kronenberg, Parvin, Rydberg, Yaswen Specific Aim 1. Determine whether rapid phenotypic responses of HMEC induced by HZE are more complex and prolonged than those induced by X-rays. Rydberg and Groesser : We have performed studies to examine the persistence of gamma-H2AX, 53BP1 and ATM 1981p foci as indication for persistent DSBs in HMEC cells. Experiments with 184v HMEC in monolayer cultures were carried out with 1 GeV/n Fe ions (LET 150 keV/µm) at doses of 0.5 Gy and 1 Gy. Such doses are small enough to allow the cells to proliferate with minimal cell cycle delays. This contrasts with most published work addressing persistent DSBs, which were carried out with doses of 40 - 80 Gy resulting in permanent G2 block. We find that 24 hr after exposure, many more gamma-H2AX foci remain after Fe irradiation than after gamma irradiation. The result is consistent with the hypothesis that Fe ion exposure causes more resistant, difficult-to-repair DSBs than low LET gamma radiation. Although this is in agreement with previous high-dose experiments in other cell lines, it was important to verify this at low doses in our proliferating cells. However, it remains an open question whether all the foci at late time-points after irradiation necessarily reflects DSBs. In contrast to ?-H2AX foci, we find a return of 53BP1 foci to background level at late time points. This discrepancy may either indicate that the gamma-H2AX foci do not represent DSBs or it could indicate that complex difficult-to-repair DSBs are nor recognized by 53BP1. Further experiments are needed to investigate this question. Costes and Barcellos-Hoff: As noted last year, we recently published observations in human fibroblasts exposed to low-LET radiation that show a clear discrepancy between RIF counts and expected number of DSB (1). To better define the relationship between physical damage and biological context in radiation-induced foci (RIF) following HZE radiation exposure, we used a well-accepted physical model of the track structure for HZE particles developed by Ponomarev and Cucinotta to simulate the production of DSB in a modeled nucleus in comparison to experimental RIF induced by HZE. In this approach, the three-dimensional space was divided into cubical pixels of the size corresponding to that of microscope image pixels. DNA was arranged into two types of intermittent bands: dense regions of DNA based on a random-walk geometry (heterochromatin), and low-density homogenous regions (euchromatin)in the simulated nucleus. The model images were then blurred to create more realistic images by applying a Gaussian filter, which takes into account the point-spread function of the microscope. We then generated nuclear images with the frequency of DSB per pixel. Applying the Gaussian blurring to these images produced images with foci-like objects, which we refer to as pseudo-foci (pRIF). The simulations yielded 1.25 DSB/µm and 0.8 pRIF/µm along 1 GeV/amu Fe

track. As expected from the theoretical predictions the simulation generated pRIF preferentially in the denser regions of DNA and randomly located along ion tracks.

Monte Carlo simulation predicted that the initial distribution of pRIF along HZE tracks would be random. In contrast, our analysis of RIF data collected at NSRL-6 and 7 observations showed some regularity in RIF spatial distribution even at the earliest time point. We also determined that RIF were more frequently located in euchromatin than predicted by modeling and occurred predominantly at the interface of euchromatin and heterochromatin for both low and high LET. We thus hypothesize that the type of chromatin organization where DSB are generated dictates the kinetic of RIF formation and may influence DNA repair.

Specific Aim 2: Determine whether persistent phenotypic responses of HMEC induced by HZE are more complex and prolonged than those induced by X-rays.

Barcellos-Hoff and Andawarra: The NSCOR invited Dr. Les Redpath of UC Irvine to spend a day discussing the applications and limitations of current human models of carcinogenic transformation by ionizing radiation. We have used the ability of human mammary epithelial cells (HMEC) to undergo acinar morphogenesis in a reconstituted basement membrane to show that radiation exposure of individual cells leads to a persistently altered phenotype in daughter cells that alter epithelial interactions and the microenvironment (2). An important implication from these studies is that the irradiated epithelial phenotype lacks critical controls imposed by the microenvironment to maintain tissue, and potentially genomic integrity. We have generated pilot data using embedded cultures for which we analyze morphology, acinar size, expression of mesenchymal markers and epithelial markers such as E-cadherin and ß-catenin using immunofluorescence and protein blot analysis. The preliminary analysis of these specimens using confocal or deconvolution microscropy indicate that HZE is as effective as sparsely ionizing radiation in eliciting the irradiated phenotype of disrupted morhogenesis. We have now determined that irradiated HMEC exposed to TGF-ß are persistently dysfunctional in a manner consistent with epithelial to mesenchymal transition (EMT). We performed studies at the NSRL-7 using 184 HMEC, which are finite lifespan post-selection HMEC from one specimen, 184. We repeated the monolayer culture experiments that were performed at NSRL-5 comparing the phenotype of cells exposed to equitoxic doses of 1 GeV Fe or gamma-rays. HMEC underwent comparable degrees of EMT in response to either densely and sparsely ionizing radiation at equitoxic doses. Additional studies will be done at LBNL to understand the molecular mechanisms that promote TGF-ß induced EMT in HMEC. Yaswen and Mukhopadhyay: As part of our effort to develop assays to distinguish differences in the persistent biological effects of high and low LET radiation, we examined the ability of HMEC to escape the first barrier to immortality, which is called stasis. Many laboratories have found that it is exceedingly rare for normal human epithelial cells to escape the mortality barrier, called agonescence (3), to generate immortal cell lines. Thus, in these experiments, fourth passage (4p) pre-stasis HMEC derived from primary cultures grown in serum-free MEGM medium were exposed to a single 2 Gy dose of X-rays. Replicate cultures of the irradiated and sham-irradiated cells in three independent experiments using HMEC from two individuals, developed patches of small actively proliferating cells in the cultures derived from irradiated cells only. Ultimately, after considerable delay, a few patches of small actively proliferating cells appeared in the control dishes as well. However, the irradiated cultures continued to proliferate after 40 additional population doublings, while the sham-irradiated cells ceased net proliferation after 15 additional population doublings. Staining for p16 protein and SA- gal activity confirmed that 2 Gy irradiation caused striking increases in the frequency and rate of appearance of p16(-) SA- gal(-) cells with long-term growth potential. Based on these data, we hypothesize that radiation, possibly acting through the generation of reactive oxygen species, causes this epigenetic change. Using this new assay, quantification of the relative biological effectiveness of high LET particles (1 GeV Fe - 1 Gy) compared to low LET reference radiation is currently being established using pre-stasis HMEC irradiated at BNL during NSRL-8.

Aim 3 Assess whether a persistent state of genomic instability is induced in the progeny of HMEC exposed to HZE or X-irradiation.

Task Progress:

Kronenberg, Sudo, Barcellos-Hoff, Gray: This year the focus has been on the assessment of genomic instability in finite lifespan 184v HMEC using multi-color whole chromosome FISH painting to address the incidence of non-clonal structural chromosome rearrangements, and centrosome staining to monitor the fraction of cells within a colony that display supernumerary centrosomes. All experiments have been performed with colonies that survive exposure to graded doses of x-rays or 1 GeV/amu Fe ions from cells irradiated as monolayers.

G-banding analysis of un-irradiated 184v cells demonstrates a stable karyotype of 46 XX, with no deviation from this chromosome complement. Chromosome painting analysis of the progeny of un-irradiated single cells also demonstrates the stability of the karyotype, with no evidence of non-clonal structural rearrangements. In contrast, both x-ray exposure (1-6 Gy) and Fe ion exposure (0.5-3 Gy) result in a substantial fraction of clones with two or more karyotypes, indicating that there is a memory of the irradiation exposure that is transmitted to the surviving daughter cells and that is expressed as a variation in the structural assortment of the chromosomes. To our knowledge, this is the first demonstration of chromosomal instability in finite-lifespan human mammary epithelial cells exposed to ionizing radiation. The transition to genomic instability has been associated with the progression from usual ductal hyperplasia to ductal carcinoma in situ in human biopsy materials (4).

Increased frequency of supernumerary centrosomes may increase the probability of abnormal segregation of chromosomes due to multipolar mitotic spindles. This latter phenotype has been found in human mammary biopsy materials with ductal carcinoma in situ, and was associated with the development of chromosomal instability (5). There is a low but detectable level of centrosomal instability in the clonal descendents of un-irradiated HMEC clones (~1% of cells in a discrete colony have 3 or more centrosomes). Assessment of radiation-induced centrosome abnormalities in 184v cells exposed to either x-rays or Fe ions indicates a dose-dependent increase in supernumerary centrosomes in the clonal progeny of irradiated cells for each radiation type. The results for both x-rays and Fe ions are similar as a function of dose. The vast majority of clones from x-irradiated cells demonstrate an increase in supernumerary centrosomes that is > 3 S.E. above the level found in the un-irradiated clones. The clonal progeny of Fe-irradiated cells also demonstrated a similar increase in the fraction of clones with increased levels of supernumerary centrosomes.

The experiments to date indicate that exposure to either x-rays or Fe ions results in the transmission of a memory of that exposure that can be expressed many cell divisions later in the descendants of single irradiated cells. Intraclonal karyotypic heterogeneity may increase the probability of development of variants that can advance the neoplastic phenotype. Thus, exposure to both sparsely and densely ionizing radiation can result in genomic instability in the progeny of irradiated human mammary epithelial cells with finite lifespan, and elicits a series of responses that have been associated with the transition to neoplasia in human biopsy material.

Specific Aim 4: Define theoretical and system biology models of HZE damage and response relevant to carcinogenesis.

Costes, Chatterjee and Boissiere: A focus of this project is to develop theoretical models for carcinogenesis. The initial approach was based on the assumption that the mutation rate represents an "initiating event" followed by clonal expansion rate for cells that carry mutation, and subsequently acquire several other mutations either directly from irradiation, spontaneous processes or through induced instability or mutator phenotype caused by an earlier mutation, etc. This approach relates to parametric fitting of data and may not be the ultimate choice as a model for carcinogenesis. The tendency in radiation biology is to model events as hierarchical, linear responses, rather than combinatorial networks. Thinking of radiation responses as assets of a network, rather than programmed responses of cells, could be used to understand the dynamic flexibility of the network. New modeling approaches will be required to represent the non-linear aspect and adaptive behavior of system biology.

To this end, we convened an informal meeting on modeling was held in Berkeley between the LBNL NSCOR and the Tufts NSCOR led by Dr. L. Hlatky, as well as Dr. R.J.M. Fry and Dr. F. Cucinnotta, to discuss how one would incorporate the effect of the microenvironment into carcinogenic risk model. The group discussed current experimental models basis and the potential for an integrated model using well-quantitated radiation responses using systems biology principles of network interconnectivity and spatial organization of cellular phenotypes within the higher order multicellular structure (6). We believe that by improving radiation carcinogenesis models in a progressive manner by incorporating more biological mechanisms in a more realistic way will help inform mechanism based modeling of risk.

Parvin and Das: In support of systems biology, algorithms have been developed for quantifying biological properties of imaging assays for both monolayer as well as embedded cell culture models. The emphasis has been on robust geometric models and improved accuracy of capturing pertinent biological properties. These computed representations are registered with a database that maps them to experimental variables and their raw representation. Additionally, a number of invariance and information preserving features are routinely computed, which can aid in constructing predictive models.

In a parallel effort, we have developed algorithms to identify the cis-regulatory elements that direct gene expression by integrating nucleotide sequence and microarray expression data sets. We have used adaptive modeling strategies to account for complex dependencies at both transcriptional and post-transcriptional levels. The central focus has been on adaptive regression splines, which are also closely related to switch-like behavior of the response function. These algorithms are currently being applied to decipher the transcriptional endpoints of TGF-ß mediated signaling pathways in irradiated cells and the elements of transcriptional regulatory logic that discriminate between monolayer or embedded culture.

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